

MyTalipapa

Public-Market Navigation & Stall-Management System

Data Flow Diagrams (Level 0–3) Entity-Relationship Diagram & Process Flowchart

Yourdon–DeMarco notation

System Design Documentation
Version 1.0 · June 2026

Contents

Contents	2
1. System Overview	3
Notation (Yourdon–DeMarco)	3
2. External Entities	4
3. Data Stores (Data Dictionary)	5
4. Entity-Relationship Diagram (ERD)	6
5. DFD Level 0 — Context Diagram	7
6. DFD Level 1 — System Decomposition	8
7. DFD Level 2 — Process 2.0 Applications & Approval	9
8. DFD Level 3 — Process 2.2 Review & Validate	11
9. DFD Level 2 — Process 4.0 Navigate & Find Stalls	12
10. DFD Level 3 — Process 4.3 Compute Walking Route	13
11. Process Flowchart — Shopper Wayfinding	15

1. System Overview

MyTalipapa is a paperless public-market platform with two purposes. For visitors it provides wayfinding — a 360° panorama tour and an augmented-reality (AR) finder that route a shopper to a specific stall. For the market it provides management — renter/contractor/admin accounts, a stall-application and approval workflow, payments, lease contracts, and notifications.

This document models the system with Data Flow Diagrams (DFDs) at four levels of detail using the Yourdon–DeMarco notation, supported by an Entity-Relationship Diagram (ERD) of the data model and a process flowchart for the shopper wayfinding journey.

Notation (Yourdon–DeMarco)

Symbol	Meaning
Circle / bubble (blue)	Process — a transformation of data, numbered (0, 1.0, 2.1, 2.2.1 ...).
Rectangle (grey)	External entity — a person or system outside the boundary that supplies or receives data.
Open parallel lines (amber)	Data store — where data rests between processes (D1–D7).
Circle (teal)	Link to a process detailed on another DFD level.
Labelled arrow	Data flow — the label names the data that moves.
ERD: box + crow's-foot lines	Entity with attributes; line ends show cardinality (one / many).
Flowchart shapes	Stadium = start/end · rectangle = action · diamond = decision.

2. External Entities

Entity	Description	Principal exchanges with the system
Shopper / Visitor	Public market visitor using wayfinding; no account required.	In: search, navigation request, QR scan. Out: route, stall info, 360° view, turn guidance.
Renter / Applicant	Registers and applies to rent a stall.	In: registration, application, payment, appeal. Out: status, rejection reason, payment notice.
Contractor	Manages assigned stalls and reviews renter applications.	In: review decision, enable/disable stall, contract dates. Out: applications queue, stall & contract records.
Admin	Market administrator overseeing accounts and records.	In: account approvals, QR generation, record queries. Out: dashboards, records, analytics.
Notification Gateway	External email (Nodemailer) and SMS (Twilio) services.	Out: email / SMS messages dispatched by the system.

3. Data Stores (Data Dictionary)

ID	Store	Representative contents	Maintained by
D1	Accounts	User records for renters, contractors, admins — credentials, role, status.	1.0
D2	Stalls	Stall number, zone, category, map coordinates, status, current tenant.	3.0, 2.0
D3	Applications	Applicant, target stall, status, rejection reason, appeal count.	2.0
D4	Payments	Application, amount, status, date of stall-rental payment.	5.0
D5	Contracts	Tenant, stall, lease start/end, contract deadline.	3.0
D6	Notifications	Email / SMS / in-app notification log.	6.0
D7	Map / Pathway Data	Stall coordinates, column/aisle/corridor geometry, navigation graph.	4.0

4. Entity-Relationship Diagram (ERD)

The ERD shows the persistent data model behind the data stores. An Account may submit many Applications, hold many Contracts, manage or occupy Stalls, and receive many Notifications; each Application targets one Stall and is settled by a Payment; each Contract covers one Stall. Crow's-foot ends denote cardinality (a single bar = one, a fork = many).

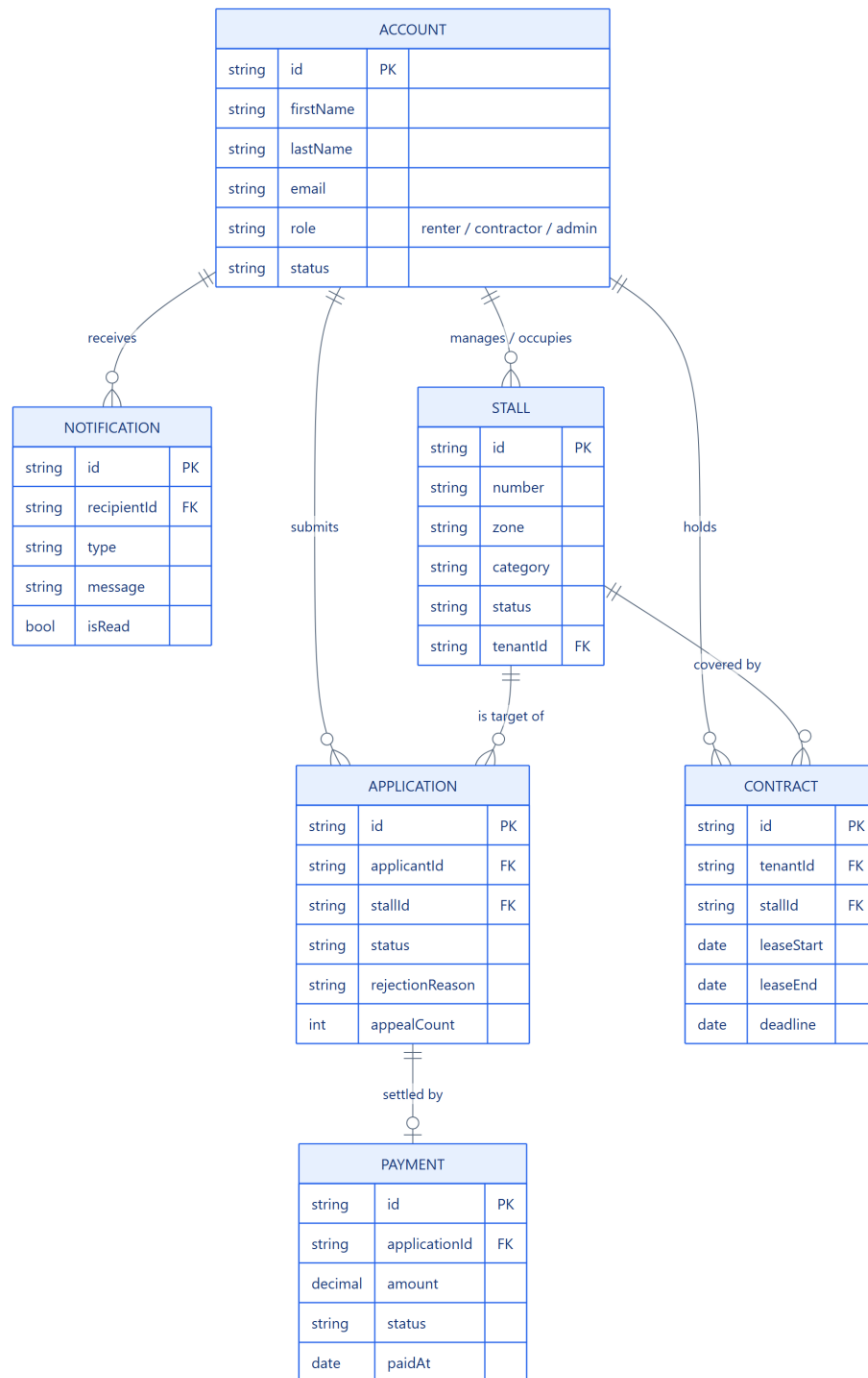


Figure 1 — Entity-Relationship Diagram

5. DFD Level 0 — Context Diagram

The context diagram shows the entire MyTalipapa system as a single process (0) and the data exchanged with each external entity. It fixes the system boundary before any internal detail is added.

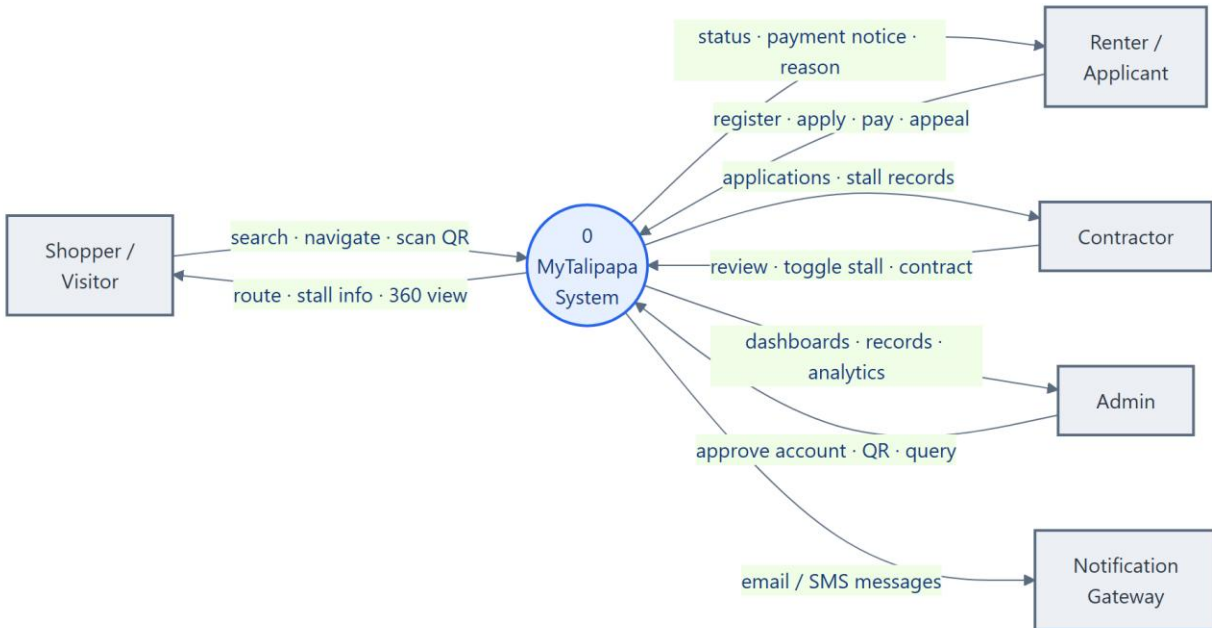


Figure 2 — Level 0 Context Diagram

6. DFD Level 1 — System Decomposition

Process 0 expands into six processes that exchange data with the entities and seven data stores.

- **1.0 Manage Accounts & Authentication** — registration, login, role-based access, admin approval of contractor accounts.
- **2.0 Process Applications & Approval** — submission, review, approve/reject with reason, appeal/resubmit, proceed-to-payment.
- **3.0 Manage Stalls & Contracts** — enable/disable stalls, maintain stall details, lease dates and deadlines.
- **4.0 Navigate & Find Stalls** — 360° panorama, search, route computation, panorama & AR guidance, QR recognition.
- **5.0 Process Payments** — record and track stall-rental payments.
- **6.0 Send Notifications** — dispatch email/SMS/in-app messages on key events.

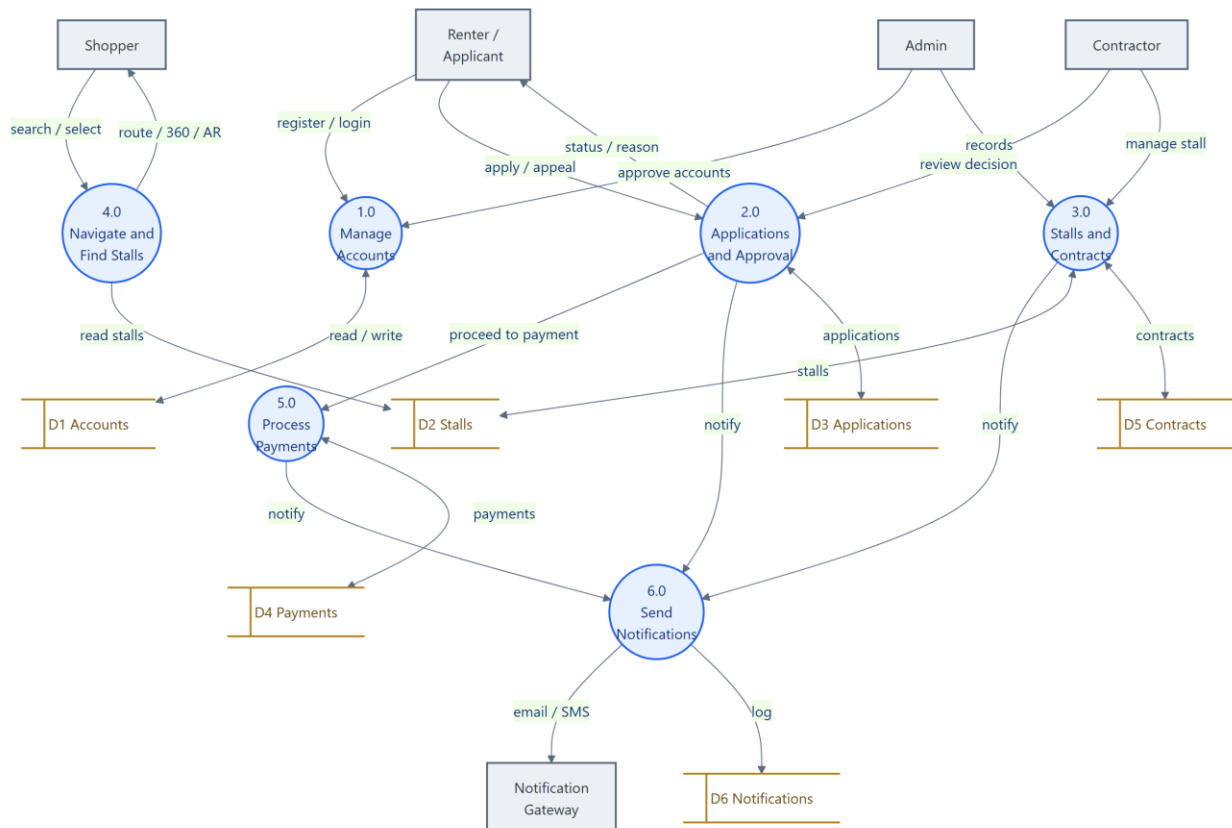


Figure 3 — Level 1 System Decomposition

7. DFD Level 2 — Process 2.0 Applications & Approval

Process 2.0 decomposes into the stall-application lifecycle: a renter submits, a reviewer validates and decides (with a reason), rejected renters may appeal, and approved applications proceed to payment and occupy the stall.

- 2.1 Submit / Resubmit Application.
- 2.2 Review & Validate (expanded at Level 3).
- 2.3 Record Decision — approve or reject, storing the reason.
- 2.4 Handle Appeal / Resubmission.
- 2.5 Initiate Payment & Occupy Stall.

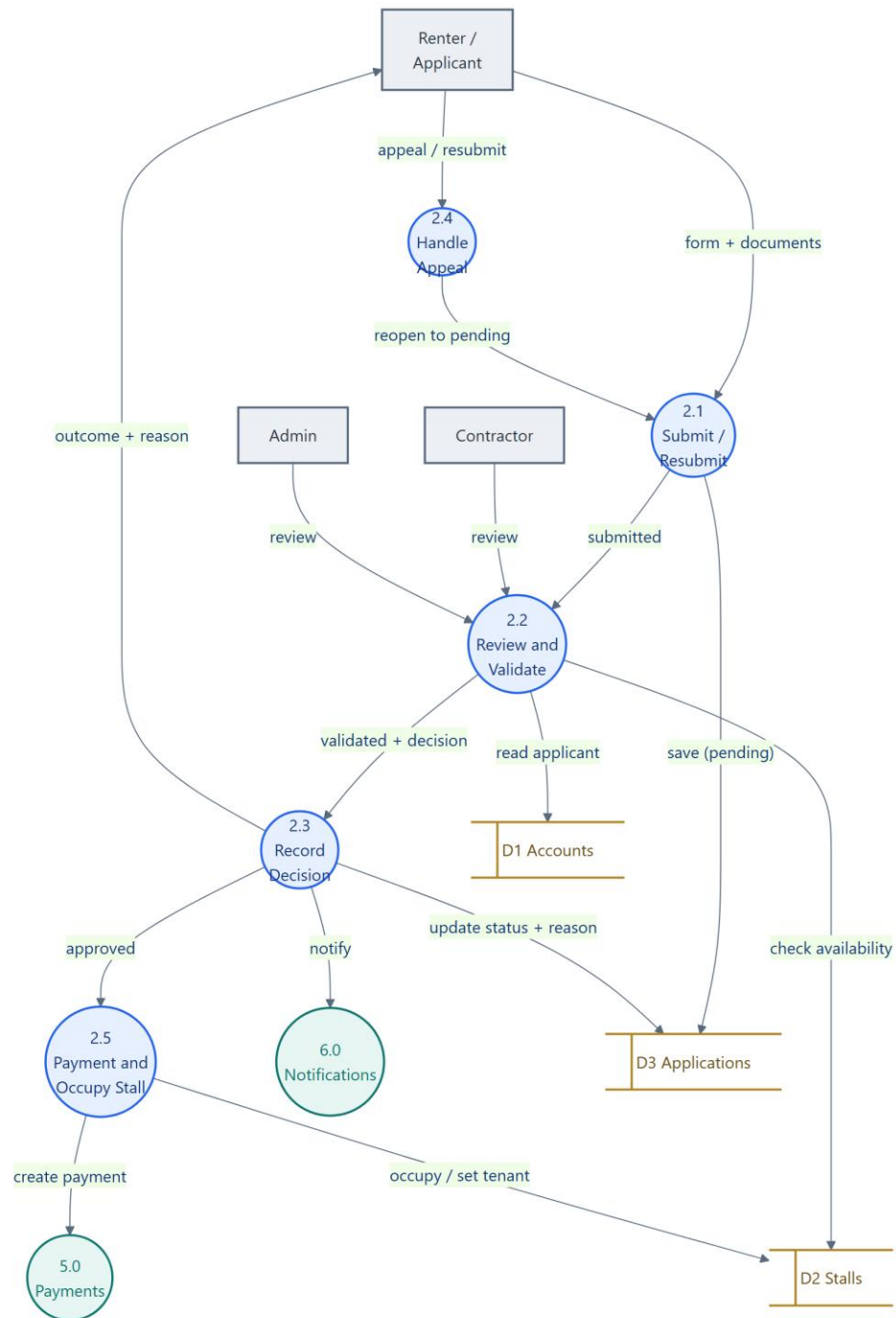


Figure 4 — Level 2: Applications & Approval

8. DFD Level 3 — Process 2.2 Review & Validate

The review process is broken into primitive steps: verifying the applicant, confirming the stall is available, presenting the case to the reviewer, and capturing the decision together with its reason.

- 2.2.1 Check Applicant Record & Documents.
- 2.2.2 Verify Stall Availability.
- 2.2.3 Present to Reviewer (contractor / admin).
- 2.2.4 Capture Decision & Reason.

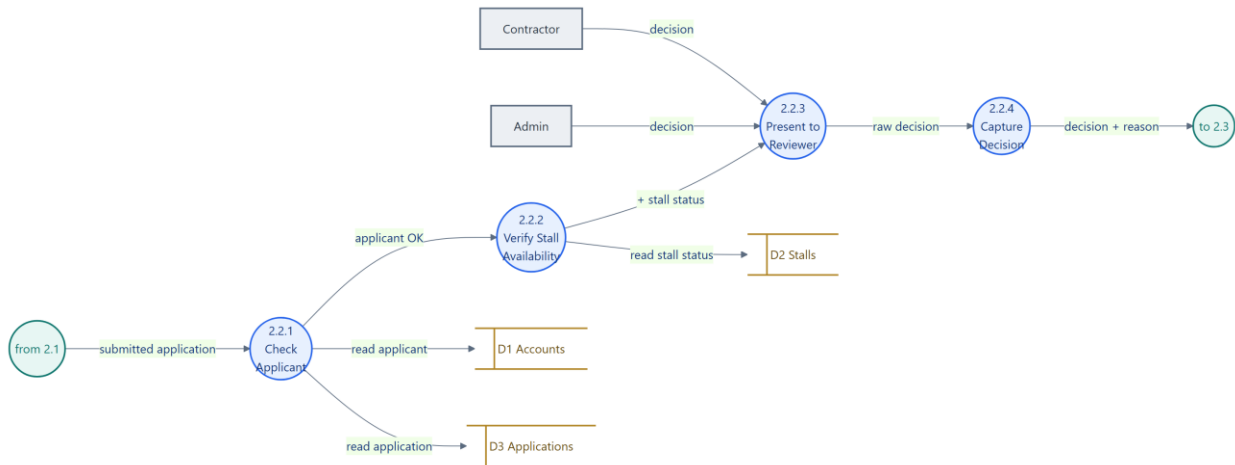


Figure 5 — Level 3: Review & Validate

9. DFD Level 2 — Process 4.0 Navigate & Find Stalls

Process 4.0 decomposes into the wayfinding pipeline: browse the panorama, choose a destination, compute a walking route, then guide the shopper either through the panorama or the AR camera.

- 4.1 Browse 360° Panorama.
- 4.2 Search / Select Destination.
- 4.3 Compute Walking Route (expanded at Level 3).
- 4.4 Guide via Panorama (arrows + mini-map).
- 4.5 Guide via AR (camera + QR + step detection).

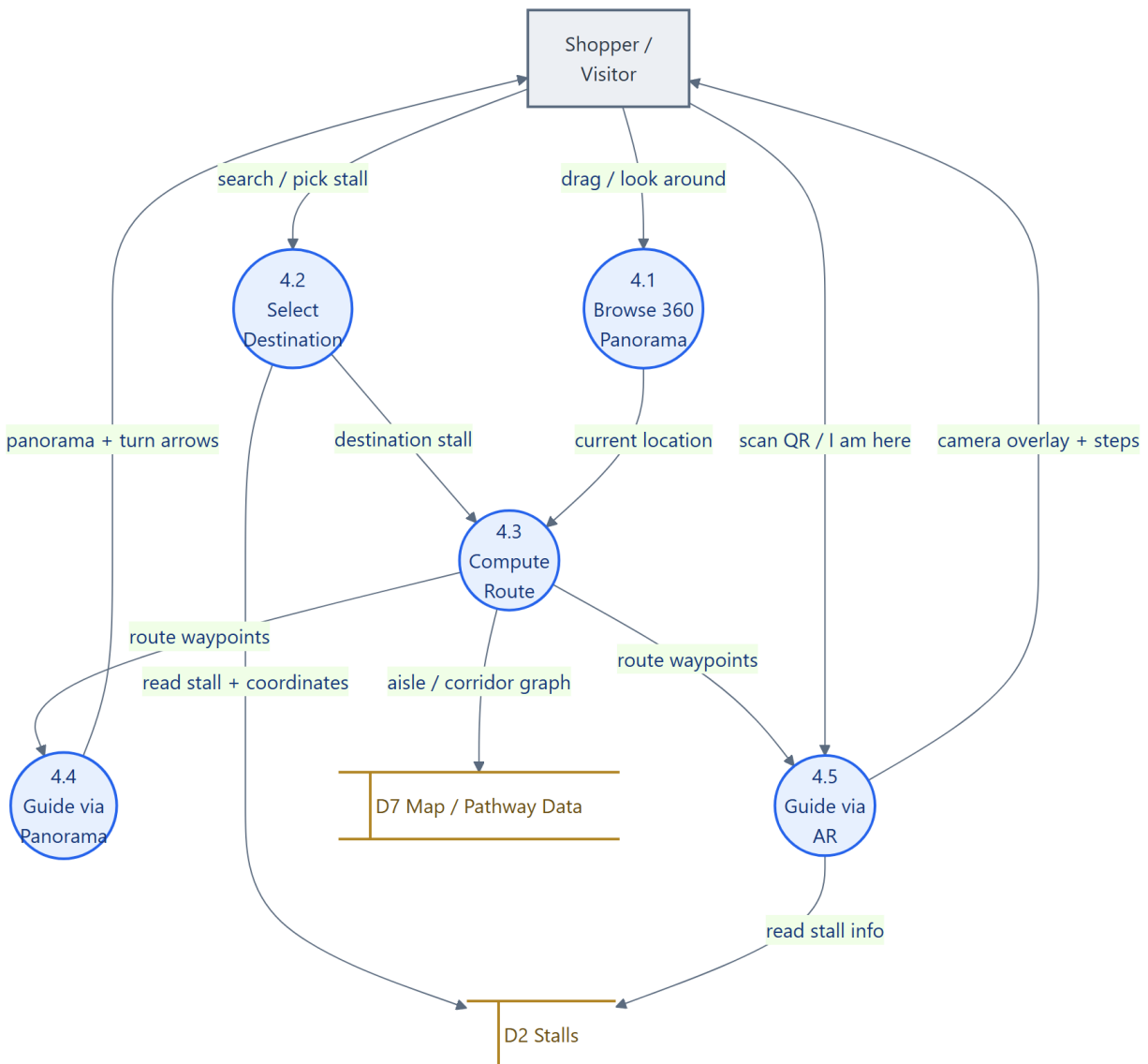


Figure 6 — Level 2: Navigation

10. DFD Level 3 — Process 4.3 Compute Walking Route

Route computation snaps the start and destination onto the nearest walkable aisles, builds an orthogonal aisle/corridor graph from the stored map geometry, runs Dijkstra's shortest-path algorithm, then simplifies the result to a minimal list of turn-by-turn waypoints. This guarantees routes follow real aisles and never cross a stall block.

- 4.3.1 Snap Endpoints onto Aisles.
- 4.3.2 Build Aisle / Corridor Graph.
- 4.3.3 Run Dijkstra Shortest Path.
- 4.3.4 Simplify to Turn-by-Turn Waypoints.

11. Process Flowchart — Shopper Wayfinding

The flowchart traces a shopper's end-to-end journey from opening the 360° tour to arriving at a stall. After a route is computed the shopper may follow panorama arrows or switch to AR; in AR they may scan a QR code to fix their position, and step detection updates their location until they arrive.

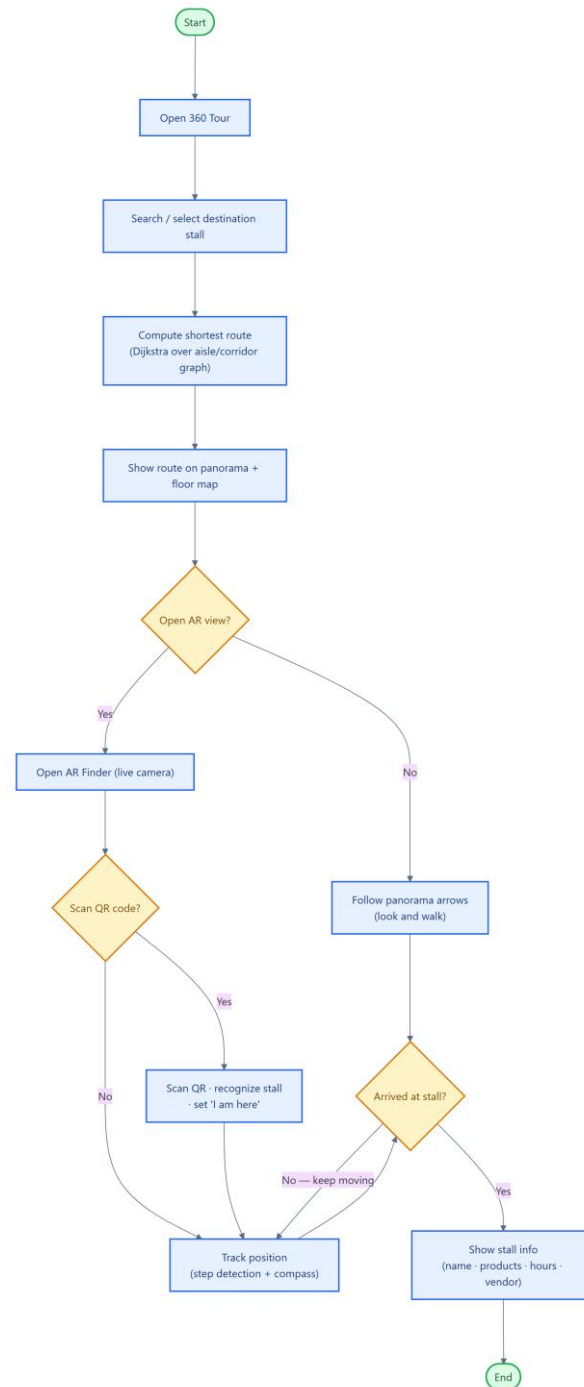


Figure 8 — Process Flowchart: Shopper Wayfinding